

FORM

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Developer's reference manual

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1 Initial remarks

This document is intended for people who are interested in understanding how FORM works internally, how to find and correct bugs in the source code, and how to extend FORM by implementing new features.

It is assumed, that the source code is available, either as a package or directly via CVS access to the FORM repository. The FORM package contains many files and several subdirectories. The actual sources of FORM, TFORM, and PARFORM are all in the directory `sources` (see section 2 for an overview). Documentation can be found in the directory `doc`. The testing suite is contained in the directory `check`.

2 Overview of the source code

Here we will discuss general aspects of the source code, i.e. the files contained in the directory `sources`.

FORM is written in ANSI C. The code is split up in header files `*.h` and source files `*.c`. Files usually don't come in pairs of a header file with the declarations and a source file with the definitions, but instead most declarations are collected in a few headers. The declaration of function headers is done in `declare.h` for example. The most prominent exceptions are `parallel.h` and `minos.h`.

Each file usually contains many hundred lines of code. To make the files more accessible, the code is structure by so-called folds. If you use the editor STedi, the code will be visualized correctly. If you use a vi-compatible editor, it is advisable to activate folds and set the foldmarkers to `set foldmarker=#[,#]`

2.1 The header files

<code>declare.h</code>	Contains the declarations of all publicly relevant functions as well as of commonly used macros like <code>NCOPY</code> or <code>LOCK</code> .
<code>form3.h</code>	Global settings and macro definitions like word size or version number. It includes several different system header files depending on the computer's architecture.
<code>fsizes.h</code>	Defines macros that determine the size and layout of FORM's internal data like the sizes of the work buffers etc.

<code>ftypes.h</code>	Contains preprocessor definitions of the codes used in the internal representation of parsed input and expressions.
<code>fwin.h</code>	Special settings for the Windows operating system.
<code>inivar.h</code>	Contains the initialization of various global data like the FORM function names or the character table for parsing. It also defines the global struct <code>A</code> , and for TFORM the struct pointer <code>AB</code> .
<code>minos.h</code>	Dedicated header to the <code>minos.c</code> source file.
<code>parallel.h</code>	Dedicated header to the <code>parallel.c</code> source file.
<code>portsignals.h</code>	Preprocessor definition of the OS signals FORM can deal with.
<code>structs.h</code>	Defines the structs that contain almost all of FORM's internal data.
<code>unix.h</code>	Special definitions for Unix-like operating systems.
<code>variable.h</code>	Some convenience preprocessor definitions to ease the access to global variables, like <code>cbuf</code> or <code>AC</code> .

2.2 The source files

<code>argument.c</code>	Code for the <code>argument</code> and <code>term</code> FORM statements.
<code>bugtool.c</code>	Low-level debugging code.
<code>checkpoint.c</code>	Code to test for checkpoint conditions, to create snapshots, and to recover from snapshot data.
<code>comexpr.c</code>	Functions the compiler calls to translate a statement that involves an algebraic expression, e.g. <code>Local</code> or <code>Id</code> .
<code>compcomm.c</code>	Functions the compiler calls to translate a statement that neither involves an algebraic expression nor is a variable declaration.
<code>compiler.c</code>	Main compiler code.
<code>compress.c</code>	Code for GZIP (de-)compression in sort files.
<code>comtool.c</code>	Utility functions for the compiler, like <code>AddRHS</code> .
<code>dollar.c</code>	Code dealing with dollar variables.

<code>execute.c</code>	Code for the execution phase of a module. Also, code dealing with brackets in FORM expressions.
<code>extcmd.c</code>	External command code.
<code>factor.c</code>	Simple factorizing code for dollar variables and expressions.
<code>findpat.c</code>	Pattern matching for symbols and dot products.
<code>function.c</code>	Pattern matching for functions.
<code>if.c</code>	Code for the <code>if</code> statement.
<code>index.c</code>	Code for bracket indexing.
<code>lus.c</code>	Code to find loops in index contractions.
<code>message.c</code>	Text output functions, like <code>MesPrint</code> or <code>PrintTerm</code> .
<code>minos.c</code>	The minos database.
<code>module.c</code>	Code for module execution and the <code>moduleoption</code> , <code>exec</code> and <code>pipe</code> statements.
<code>mpi2.c</code>	MPI2 code for PARFORM.
<code>mpi.c</code>	MPI1 code for PARFORM.
<code>names.c</code>	Name administration code to deal with the declaration of FORM variables.
<code>normal.c</code>	Code to normalize terms, i.e. bring them to standard form.
<code>opera.c</code>	Code for doing traces, contractions, and tensor conversions.
<code>optim.c</code>	Code to optimize FORTRAN or C output.
<code>parallel.c</code>	PARFORM (MPI-independant code).
<code>pattern.c</code>	General pattern matching and substitution.
<code>poly.c</code>	Code for polynomial arithmetic (experimental).
<code>polynito.c</code>	Code for polynomial arithmetic and manipulation.
<code>pre.c</code>	The preprocessor.
<code>proces.c</code>	The central processor.
<code>ratio.c</code>	Partial fractioning and summing functions.
<code>reken.c</code>	Code for numerics.
<code>reshuf.c</code>	Utility functions for the renumbering of dummy indices, and for statements like <code>shuffle</code> , <code>stuffle</code> , <code>multiply</code> .
<code>sch.c</code>	Code for the textual output of terms and expressions.

<code>setfile.c</code>	Code to deal with setup parameters and setup files.
<code>smart.c</code>	Code doing optimized pattern matching.
<code>sort.c</code>	Code for the sorting of expressions.
<code>startup.c</code>	Start of program (<code>main()</code>). Code for the startup and shutdown phase of FORM.
<code>store.c</code>	Code to read from disk or write to disk terms and expressions. Also, store file and save file management.
<code>symmetr.c</code>	Pattern matching for functions with symmetric properties.
<code>tables.c</code>	Code for the tablebases.
<code>threads.c</code>	TFORM. Almost all of the TFORM specific code.
<code>token.c</code>	The tokenizer.
<code>tools.c</code>	Utility functions to deal with streams, files, strings, memory management, and timers.
<code>unixfile.c</code>	Wrapper functions for UNIX file I/O functions.
<code>wildcard.c</code>	Code for wildcards.

2.3 The global structs

FORM keeps its data organized in several global structs. These structs are defined in `structs.h` (in the fold A) and come by the names `M_const`, `P_const`, The various global variables are grouped in these structs according to their rôle in the program. The fold commentaries give details on this. `M_const` is for global settings at startup and `.clear`, for example.

The various structs are collected in the struct `AllGlobals`. In the case of sequential FORM, this struct is made into the type `ALLGLOBALS`, and in `inivar.h`, the global variable `A` is defined having this type. This global variable `A` holds all the data defined in the various structs. In `variable.h` several macros are defined to simplify (and more importantly unify) the access to the struct elements. For example, one can access the variable `S0` in `T_const` as `AT.S0`.

With the multi-threaded version TFORM things are a little bit more complicated, because some data needs to be replicated and made private for each thread. This kind of data is situated in the structs `N_const`, `R_const`, and `T_const`. For TFORM, these structs are collected in the struct `AllPrivates` (which makes up the type `ALLPRIVATES`), all other structs go into the `AllGlobals` struct. The global variable `A` now contains only the

non-thread specific data. For each thread a `AllPrivates` struct is dynamically allocated and the global pointer variable (in `inivar.h`) `AB` holds their references. `AB` is an array of pointers where the index corresponds to the thread number. The macros defined in `variable.h` to access the global struct data are made such that they transparently work with the `AB` array. The user doesn't need to care about these details and can still write as in the previous example `AT.S0`. This keeps the code of sequential FORM and multi-threaded TFORM uniform.

The only small price one has to pay to make this uniform access by macros possible is to make sure every function in FORM knows in which thread it is executed. The `AN`, `AR`, and `AT` macros use a variable `B`, which is set to the correct entry in `AB` by one of two ways. First, a function can use the macro `GETIDENTITY` (defined in `declare.h`). In TFORM it calls `WhoAmI()` to get the thread number, declares the pointer `B`, and sets `B` to point to the correct entry in `AB`. In sequential FORM this macro is empty. The second way is to get the variable `B` as a parameter from the caller. For this method the macros `PHEAD`, `PHEAD0`, `BHEAD`, and `BHEAD0` exist (defined in `ftypes.h`), which can be used in the parameter list of the function declarations. The variants with a zero differ only by not including a trailing comma, which is not allowed if no other parameters are following in the declaration. Usually, `PHEAD` is used in the declaration (it includes type information), while `BHEAD` appears in the calling of functions. Which way to set `B` is chosen, depends on the use of the function. The `PHEAD` method is faster than `GETIDENTITY` and should be preferred in functions that are called very often. On the other hand, `GETIDENTITY` is more general as it does not rely on every caller to supply `B`.

The elements of the structs are of various types. Some types are just simple macros mapping directly to built-in types (see `form3.h`) like `WORD`, others are names for structs that are defined (mostly) in `structs.h`. Often, variables of the same type are grouped together to help the compiler with alignment. Also, a lot of structs use macros like `PADLONG` (`unix.h` or `fwin.h`) to pad a struct such that its size is a multiple of a built-in type size. This again is to help with the data alignment.

Most struct elements have comments that explain their use. These commentaries often include the information where this element was once located in the old version 2 of FORM (it is the pair of parentheses with or without a capital letter inside). Pointers come in two flavors: Some pointers reference a dynamically allocated piece of memory, basically owning this memory. Others just reference another variable or point into allocated memory. The first kind is usually marked with `[D]` for easy identification. These point-

ers often need to be treated particularly, e.g. during the snapshot creation, when recovering, or when shutting down.

During start up (`main()`), all the memory of these global structs, i.e. their element variables, is initialized to zero.

2.4 Configuration

The source code evaluates several preprocessor definitions that can be defined by the user. According to these definitions the executable can be configured in different ways. As a default, the sequential version of FORM is generated. But if, for example, the preprocessor variable `WITHPTHREADS` is defined, the multi-threaded version `TFORM` will be compiled. These preprocessor variables can be set when calling the compiler, like

```
gcc -c -DWITHPTHREADS -o pre.o pre.c
```

The most commonly considered preprocessor variables are: `WITHPTHREADS`, `PARALLEL`, `WITHZLIB`, `WITHGMP`, `WITHSORTBOTS`, `LINUX`, `OPTERON`, `DEBUGGING`. The first two change the flavor of the executable: `TFORM` or `PARFORM`. The next two configure whether FORM uses the `zlib` library for compression during sorts or the `GMP` library for arbitrary precision arithmetics. The next decides whether FORM uses dedicated sorting threads in `TFORM`. `LINUX` specifies that the executable is to be compiled for a Linux or UNIX compliant operating system. An alternative here would be to set the variable `ALPHA` or `MYWIN64` instead, but these builds are less common. `OPTERON` has to be set if one compiles a 64bit executable. `DEBUGGING` enables some features for a non-release debugging version of the executable (commonly named `vorm` or `tvorm`).

When using the `autoconf` setup, the settings concerning the operating system, architecture (32/64bit), and flavor of the executable are automatically done right. Additional settings like `WITHZLIB` can be changed by manually editing the file `config.h`, which is included in `form3.h`.

Version numbers and production date can also be set, but then one either needs to edit the appropriate lines in `form3.h` when in a manual compiling setup, or by editing `configure.ac` in an `autoconf` setup.

3 Discussion of a typical FORM run

We discuss in the following what is happening inside FORM when it executes a given program. The discussion focuses more on the interplay between the various parts of FORM and on key concepts of the internal data representation than on in-depth details of the code. For the latter, the reader is

referred to section 4. This section should for better comprehension be read with the referenced FORM source files opened aside.

We consider the following exemplary FORM program `test.frm` (which we run with the command `"form test"`):

```

1      #define N "3"
2
3      Symbol x, y, z;
4
5      L f = (x+y)^2 - (x+z)^(N);
6      L g = f - x;
7
8      Brackets x;
9      Print;
10     .sort
11
12     #do i=2,3
13     Id x?^'i' = x;
14     #enddo
15
16     Print +s;
17     .end

```

The entry function `main()` is in `startup.c`. It does various initializations before it calls the preprocessor `PreProcessor()`, which actually deals with the FORM program. The code shows some typical features: Preprocessor macros are frequently used to select code specific to certain configurations. The two most common macros can be seen here: `WITHPTHREADS` for a TFORM executable and `PARALLEL` for a PARFORM executable. Macros are used to access the global data contained in the variable `A`, like `AX.timeout` for example. The code uses (usually) own functions instead of standard functions provided by the C library for common tasks. Examples in `main()` are `strDup1` or `MesPrint` (replacing `printf()`). Another very often used function is `Malloc1()` replacing `malloc()`. The reasons are better portability and the inclusion of special features. `Malloc1()` for example makes a custom memory debugger available while `MesPrint()` knows among other things how to print encoded expressions from the internal buffers.

The initializations in `main()` are done in several steps. Some like the initialization of `A` with zeros is done directly, most others are done by calls to dedicated functions. The initializations are split up according to the type of

objects involved and the available information at this point. The command line parameters passed to FORM (none in our example run) are treated in the function `DoTail()`. After that, files are opened and also parsed for additional settings. Then, as all settings are known, the large part of the internal data is allocated and initialized. Finally, recovery settings are checked, threads are started if necessary, timers are started, and variable initializations that might need to be repeated later (e.g. clear modules) are done in `IniVars()`.

The call to `OpenInput()` reads the actual FORM program into memory. The input is handled in an abstract fashion as character streams. The stream implementation (`tools.c`) offers several functions to open, close, and read from a stream. Streams can be of different types including files, in-memory data like parts of other streams or dollar variables, as well as external channels. The access to the characters in all streams though is nicely uniform. In `OpenInput()` a stream is representing our input file. Most of the logic there deals with the jump to the requested module (skipping clear instructions). It uses the function `GetInput()` to get the next character in the stream. Which stream it reads from is determined by the variable `AC.CurrentStream`. This global variable in the sub-struct `C.const` of the `ALLGLOBALS` variable `A` is an example of how the different parts of FORM typically communicate with each other by means of global variables.

Next is the preprocessor. The preprocessor is implemented in the function `PreProcessor()` in `pre.c`. This function consists basically of two nested for-loops without conditions (`for (;;) { ...}`). The outer loop deals with one FORM module for each iteration, the inner loop deals with one input line. We have certain initializations done before in our example the code runs into the inner loop, where `GetInput()` reads our input file. The variables are all set such that the reading starts from the beginning of our input file.

The input in variable `c` is tested for special cases. Whitespaces are skipped. Comments starting with a star `*` (unless `AP.ComChar` is set to a different character) are also skipped including whole folds. The crucial check on `c` is the if-clause that checks it for being a preprocessor command (`#`), a module statement (`.`), or something else which is usually an ordinary statement.

```
1      #define N "3"
```

In our case, we have a preprocessor command in the input. The function `PreProInstruction()` is called to read and interpret the rest of the line.

The first part deals with the loading of the command in a dedicated buffer. For the moment, we ignore the details for the special treatment of cases when we are already inside a if or switch clause in a FORM program. In our run, the function `LoadInstruction()` is simply called.

`LoadInstruction()` copies input into the preprocessor instruction buffer. Three variables govern this buffer: `AP.preStart` points to the start of the buffer, `AP.preFill` to the point where new input can be copied to, and `AP.preStop` to (roughly) the end of the buffer. This setup is quite typical for buffers in FORM. The memory is allocated at the start of FORM. Later, like at the end of `LoadInstruction()`, if the buffer gets too small, it can be replaced by a larger memory patch with the help of utility functions like `DoubleLList()`. The contents is copied from the old to the new buffer. Since this dynamical resizing of buffers needs to be done with most buffers occasionally, most buffers in FORM store data such that it easily allows for copying, i.e. usually C pointers are avoided and instead numbers representing offsets are used. Since the preprocessor instruction buffer just contains characters there is no problem here.

In `LoadInstruction()` with our input and the mode set to 5 the input is just copied directly without any special actions taking except for a zero that is added at the end of the data. `PreProInstruction()` examines the data in the preprocessor instruction buffer for special cases, and then does a look-up in the `precommands` variable. This is a vector of type `KEYWORD` which enables the translation of a string (the command) to a function pointer (the C function that performs the operations requested by preprocessor command). `FindKeyword()` does these translations and the found function pointer is then dereferenced with the rest of the input in the instruction buffer as an argument.

The function pointer will point to `DoDefine()` in our case. `DoDefine()` just calls `TheDefine()` that does the work. The if-clauses for `AP.PreSwitchModes` and `AP.PreIfStack` are present in most of the functions dealing with preprocessor commands. They check whether we are in a preprocessor if or switch block that is not to be considered, because the condition didn't hold. Then, the standard action is to just exit the current function leaving it with no effect. Since there are preprocessor commands like `#else` or `#endif` this decision can only be taken at this level of the execution and requires the repeated use of this idiom.

The function scans through possible arguments and the value. In the value, special characters are interpreted. Ultimately, the preprocessor variable is created and assigned in the called function `PutPreVar()`. The variable `chartype` deserves an explanation. One will find it used very often in

the C code that does input parsing. `chartype` is actually a macro standing in for `FG.cTable`. This global, statically initialized (in `inivar.h`) vector contains a value of every possible ASCII character describing its parsing type. The parsing type groups different ASCII characters such that the syntax checking is facilitated, see `inivar.h` for details.

In `PutPreVar()` we get into the details of the name administration. We will just comment on some of the more general features. `NumPre` and `PreVar` are macros to access elements in `AP.PreVarList`. The type of `AP.PreVarList` is `LIST`. This is a generic type for all kinds of lists and it is used for many other variables in FORM. A `LIST` stores list entries in a piece of dynamically allocated memory that has no defined type (`void *`). The utility functions for managing `LIST`s like `FromList()` are ignorant about the actual contents and perform list-specific operations like adding, removing or resizing a list. An actual entry can be accessed by some pointer arithmetic and type casting. The `PreVar` macro contains such a cast to the type `PREVAR` which represents a preprocessor variable.

`PutPreVar()` creates a new list entry for us and basically copies the contents of the parameter `value` to the memory allocated to `PREVAR`'s `name`. So, by writing `PreVar[0]->name` or `PreVar[0]->value` we could access the strings `N` or `3`.

In `TheDefine()` the function `Terminate()` is used several times. This function ultimately exits the program, but first tries to clean up things and print information about the problems causing this program termination.

```
2
3      Symbol x, y, z;
```

In our run, we return to the function `PreProcessor()` and start a new inner loop iteration that reads a new line. After skipping the empty line we end up in the else-branch of the big if-clause testing `c` this time. Here the major steps are: we check again whether we are in a preprocessor if or switch, call `LoadStatement()` to read and prepare the input, and call `CompileStatement()` to perform the actions requested by the statement. The program enters the compiler stage.

We also see a call to `UngetChar()`, which puts back the character that has been read into the input stream. This is necessary, because `LoadStatement()` and `CompileStatement()` need the complete line for parsing. The variable `AP.PreContinuation` is used several times. This variable deals with statements that span several input lines. `LoadStatement()` can recognize unfinished statements and sets this variable accordingly.

`LoadStatement()` basically copies the input to the compiler's input buffer at `AC.iBuffer` (which has `AC.iPointer` and `AC.iStop` associated to it). It modifies the copy if necessary. The modification are to replace spaces by commas or insert commas at the right spots to separate tokens. The interpretation steps that are following rely on these syntactic conventions.

The call to `CompileStatement()` is done only if no errors occurred and all lines of a statement have been gathered into the compiler's input buffer. `CompileStatement()` is called with the address of this input buffer and tries to identify the statement. Like in the preprocessor, the input string is search in a vector of `KEYWORDS` (in `compiler.c` and if found, a function pointer is dereferenced to the function that actually deals with the command and its options and arguments. Here, we have actually two vectors of `KEYWORDS`, because some statements might be stated in abbreviated form. The function `findcommand()` deals with the search. `CompileStatement()` does some small extra work, like for example checking the correct order of statements. In our case, it calls the function `CoSymbol()`. This functions is in file `name.c`, because as a declaration it basically adds something to the name administration. Functions for other statements can be found in `compcomm.c` and `compexpr.c`.

`CoSymbol()` loops over the arguments and adds proper variable names together with their options to the symbols list `AC.Symbols` and the name administration (in the call to `AddSymbol()`). In our case, we have `x`, `y`, and `z` added. We have already encountered the basic mechanism of how a specific struct is added to a `LIST`. The name administration was not explained before, though.

Symbols can appear in expressions that need to be encoded. The coding for symbols can simply be its entry index in the list `AC.Symbols`, but symbols also need to be recognized when an expression is parsed. Therefore a efficient look-up mechanism is required. This is achieved by a second data structure that holds the name strings in a tree for fast searching. The data in the symbol list does not contain the name string itself, but contains a reference (a index) into this name string tree. The tree is managed by generalized functions and types that are also used for other, similar objects like vectors, indices, etc. The functions for name trees are located in the first part of the file `name.c`. The types `NAMENODE` and `NAMETREE` are defined in `structs.h`. `NAMENODES` are the node of a balanced binary tree. It does not hold the name string just an index into `NAMETREE`. The actual data is contained in `NAMETREE` that constitute one tree. This type has buffers for the nodes and for the name strings. This has the benefit of avoiding small malloc calls for individual nodes. Also, since all referencing is done via offsets into

these buffers, a relocation or serialization of such a tree is very easy. In the struct `C_const` (aka the global AC) several name trees are defined, for dollar variables, expressions, etc. The symbols added in our example program go into the nametree referenced by `AC.activenames`, which is at this point equal to `AC.varnames`.

Our program returns to the `PreProcessor()` and starts parsing the next lines:

```
5      L f = (x+y)^2 - (x+z)^'N';
6      L g = f - x;
```

This time the function `DoExpr()` will get called (via `CoLocal()`) for each line to do the parsing. The function `DoExpr()` first tries to figure out what type of `Local` statement we have. In our cases we have an actual assignment. With the call to `GetVar()` we check whether a variable of the same name already exists. The search is done in the nametrees `AC.varnames` and `AC.exprnames`. Since our names are new we don't find a previous variable and simply call `EntVar()`. `EntVar()` creates an entry in `AC.ExpressionList` and puts the name into the `AC.exprnames` name-tree. The entry in `AC.ExpressionList` is of type struct `ExprEsSiOn`. There are more struct elements than in the case of symbols, but the principle is the same. Up to now, the right-hand-side (RHS) has not been looked at and therefore no information about it is saved in the expression's entry yet. The connection between the expression's entry in the `AC.ExpressionList` and the data containing the RHS will be made via the elements `prototype` and `onfile` as we will describe soon. The access to elements in `AC.ExpressionList` is facilitated by the macro `Expressions`. The following code in `DoExpr()` builds up a so-called prototype and puts the RHS in encoded form into the buffer system via the call to `CompileAlgebra()`.

FORM uses the allocated memory in `AT.WorkSpace` for operations like the generation of terms. This memory stores `WORDS` and is used in a stack-like fashion with the help of the pointer `AT.WorkPointer`. A function can write to this memory and set `AT.WorkPointer` beyond the written data to insure that other functions that are called and might use the workspace as well do not overwrite this data. It is the responsibility of the function to reset `AT.WorkPointer` to its original value again (see variable `OldWork` in our case). Every thread in TFORM will have its own private work space.

FORM now uses `AT.WorkSpace` to build up a data structure that contains everything that needs to be known at a later stage about the expression that is parsed. The creation and the layout of the data is quite typical. First

comes a header that signifies what is coming. Here, it is `TYPEEXPRESSION`. Then comes the length of the whole data, i.e. the total number of occupied `WORDS`. The actual contents is following, which is a so-called subexpression that we will discuss soon. The contents is followed by a coefficient and a zero, which signifies the end of the data.

Coefficients are coded in FORM always in the following manner: Since coefficients can in general be fractional numbers, we encode an integer numerator and an integer denominator. The integers can have arbitrary length (limited only by the buffer sizes, see the setup variables `MaxNumberSize` and `MaxTermSize`) and are encoded in `WORD`-pieces in little-endian convention. The number of allocated `WORDS` is always the same for the numerator and the denominator. The last word of the coefficient contains the size of the whole coefficient in words. The formal structure of a coefficients is therefore like this:

NUMERATOR WORDS, DENOMINATOR WORDS, LENGTH.

The integers are always unsigned, i.e. positive. Negative fractions are encoded by a negative length. Examples (with 16bit words): $2^{16} + 2 = 65538$ gives words 2,1,1,0,5 and $-5/2$ gives 5, 2, -3.

The data structure in `AT.WorkSpace` is basically an instruction for the generator, a central function that does the main work during the execution of the FORM program, to generate an expression. The content of the expression is a subexpression. This is a pointer to the real content of the expression and will be substituted later after the execution. The main reason for this delayed expression insertion is that it can often save a lot of intermediate operations and data space and thereby speed up FORM. A case where such a thing can happen is, when an expression is used at different places and the different parts are brought together by some operations. Then, cancellations may occur or terms can be factored out and when the expressions finally is inserted the workload is less.

In our example run, the data that will later instruct the generator to create an expression looks in total like this:

*TYPEEXPRESSION, SUBEXPSIZE+3, 9, SUBEXPRESSION,
SUBEXPSIZE, 0, 1, AC.cbufnum, 1, 1, 3, 0*

We used the macro names as in the actual code. `AC.cbufnum` is a variable that is the index of the compile buffer used for this parsed statement. At the end of the data preparation phase the pointer `AT.WorkPointer` is set

beyond the data on the trailing zero, the pointer `AT.ProtoType`, which is used soon in following functions is set to the word `SUBEXPRESSION`.

The expression will be put into the scratch buffer system. This system comprises the small and large buffers and the scratch files. Where new data to the scratch buffers will be stored is of no concern to a function like `DoExpr()`, it simply uses several utility functions for that purpose. Still, we need to initialize the variable `pos` here that will indicate the position of the data, i.e. the expression, in the scratch file.

Next, the function `CompileAlgebra()` is called to parse the right hand side and put the codified expression into the FORM buffers. It basically calls two functions: `tokenize` and `CompileSubExpressions`. `tokenize` is the tokenizer that translates the input character string in a sanitized and partly interpreted string of codes. It will look up the variables named in the input string and put the index they have in the name administration into the tokenized output. Our input string is transformed into the code string like this

```
(      -13  LPARENTHESIS
x      -1   TSYMBOL
      5
+     -26   TPLUS
y      -1   TSYMBOL
      6
)     -14   RPARENTHESIS
^     -25   TPOWER
2      -8   TNUMBER
      2
-     -27   TMINUS
(      -13   LPARENTHESIS
x      -1   TSYMBOL
      5
+     -26   TPLUS
z      -1   TSYMBOL
      7
)     -14   RPARENTHESIS
^     -25   TPOWER
'N'    -8   TNUMBER
      3
      -29   TENDOFIT
```

This code string then lies in the `AC.tokens` buffer where it is used by

subsequent functions.

The function `CompileSubExpression()` finds terms in an expression that might be reused at another place and extracts them. As one can see in the code, the function looks for terms in parentheses and works recursively. The end of such a term is each time marked with `TENDOFIT`. Then, the function `CodeGenerator()` called at the end of `CompileSubExpression()` does the real work.

In our example `CodeGenerator()` first gets the data

LPARENTHESIS, TSYMBOL, 5, TPLUS, TSYMBOL, 6, TENDOFIT

as a parameter, which is the term $x + y$. It builds up the actual term encoding in the workspace and first reserves for that enough space there. One can see the pointer arithmetic using constants like `AM.MaxTal`, which is the maximum number of words a number can occupy. It reserves space for the coefficient, an integer, and the actual term. Once a token is recognized, the equivalent term data is written to the workspace and the function `CompleteTerm` is called. This function completes the data to

8, 1, 4, 5, 1, 1, 1, 3, 0.

The first word is the total length, i.e. 8 words. This is the length of the whole expression. The second word is the type of the term, which is a symbol. It is the value `SYMBOL` as defined in `ftypes.h`. This macro definition `SYMBOL` has the value 1 (in the FORM version at this time this reference is written). Following the type signifying word is the length of the term, which is 4. Several such terms could follow each other, but we only have one term at the moment. Finally, we have the trailing words for the coefficient being 1 and a terminating zero. The meaning and interpretation of the words in the data of a single term after the type word and the length word are dependent on the type. For symbols, we have pairs of word, where the first word is the index of the symbol in the name administration and the second word is the exponent. Here we have symbol 5 ($= x$) with an exponent 1. After `CompleteTerm()` has constructed the whole expression it copies the data to the compile buffers with the help of the function `AddNtoC()`.

The compile buffers contain the instruction for the execution engine, the `Processor()`, that will start when the `.sort` command is parsed. Our terms are put into the right-hand-side buffers in the compile buffer. When the `Processor()` will read these buffers one after the other, it will take the terms and put them into the scratch buffer system. Then, they become the expressions upon which further statements do act. The compile buffers are

stored in the list `AC.cbufList` and we get access to the elements via the cast `((CBUF *) (AC.cbufList.lijst))`. This cast is defined as a preprocessor macro called `cbuf`. The element `cbuf[0]->numrhs` (0 is the current compile buffer we are using) gives the number of entries in `cbuf[0]->rhs`, which is an array of pointer into `cbuf[0]->Buffer`. We have 3 elements:

```
cbuf[0]->rhs[1]    -->
    8, 1, 4, 5, 1, 1, 1, 3, 8, 1, 4, 6, 1, 1, 1, 3, 0
cbuf[0]->rhs[2]    -->
    8, 1, 4, 5, 1, 1, 1, 3, 8, 1, 4, 7, 1, 1, 1, 3, 0
cbuf[0]->rhs[3]    -->
    9, 6, 5, 1, 2, 0, 1, 1, 3, 9, 6, 5, 2, 3, 0, 1, 1, -3, 0
```

`cbuf[0]->rhs[0]` is not used and the data lies consecutively in `cbuf[0]->Buffer`. The meaning of the first two entries has already been explained. These are expressions containing $x + y$ and $x + z$, respectively. The last expression uses subexpressions that have the type `SUBEXPRESSION` = 6. The length of a subexpression is 5 and the contents 1,2,0 means that expression 1 needs to be inserted with an exponent of 2. The zero is a dirty flag that signals to the processor the state of the subexpression. Here in the compile buffers it is simply cleared to zero. The contents 2,3,0 of the second subexpression should be obvious. Finally, we have an negative coefficient for the second subexpression which accounts for the minus sign between the parentheses in our original expression.

We return to the function `DoExpr()` where the prototype of the expression is put into the scratch system via the call `PutOut()` and we are finished with this line in the input file. The next line defining a second local expression works the same.

We come to the parsing of the following statements:

```
7
8     Brackets x;
9     Print;
```

The bracket statement is dealt with in function `DoBrackets()`. It sets the flag `AR.BracketOn` to 1 and constructs the term that will stand outside the bracket. This term is copied into the `AT.BrackBuf` buffer, where it can be used by the execution engine when it needs to insert this heading term into an expression.

The print statement is parsed in function `DoPrint()`. Since we don't have any arguments to `Print` all active expressions shall be printed. `DoPrint()`

just loops through the **Expressions** list and sets the **printflag** to 1 for each expression.

With the next statement in our input file

```
10      .sort
```

we will get to know the other central parts of FORM: the processor and the sorting routines. The code in the **PreProcessor()** will call **ExecModule()** which calls **DoExecute()**. We can ignore a lot of code there that is only for parallelized versions of FORM. There are three important functions calls happening. First, **RevertScratch()** is called. FORM uses three scratch buffers: input buffer, output buffer, and the hide buffer. The usual mode of operation is to apply statements on expressions in the input buffer, sort and normalize the result, and write it into the output buffer. This repeats for every executing module and therefore an important optimization is made: the input buffer and the output buffer simply change their roles. **RevertScratch()** does this job. The second and third important calls are to **Processor()** and **WriteAll()**.

Processor() is, as the name suggests, the main processor that executes statements and deals with the results. A lot of initialization work is done before we go into the large loop over the expressions that spans almost the whole function. Our expressions have as regular expressions from the scratch buffers the **inmem** flag set to zero, so we go into the else branch of the checking if-clause. There we go to the case of a **LOCALEXPRESSION**. The main logic here is to do a single call to **GetTerm()** to get the first term from the input file and copy that to the output with the call to **PutOut()**. This first term, which is a subexpression, serves as a header for the expression. It follows a (while-)loop that calls **GetTerm()**, and if there are still terms, the loop executes its body and calls **Generator()**. After this loop, some clean-up and a final **EndSort()** is done, before the outer loop over the expressions repeats. **Generator()** is the function where the read input, which is *9, 6, 5, 3, 1, 0, 1, 1, 3*, will be substituted and expanded.

Generator() gets the term in the workspace and first tries to do all substitutions (**SUBEXPRESSION**), then applies the statements in the compile buffers to the normalized terms, substitutes again if necessary, do brackets, and finally sorts the result.

The call to **TestSub()** does the search for subexpressions. **TestSub()** will find a subexpression in our case and return the number (3) of this subexpression and set other global variables ready for the following steps. In **Generator()** we enter therefore the if-clause checking **replac** > 0. Depending on the power of the subexpression different operations are taken.

We have our subexpression to the power one only, which is an easy case. The actual substitution is performed by the function `InsertTerm()`. Since the new term might again contain subexpressions we do a recursive call to `Generator()`. Our expression contains several layers of subexpressions which are all dealt with as described above. Only the powers of the other subexpressions are different from one, so we get slightly more work to be done which involves the expansion of the terms using binomials.

Finally, the call to `TestSub()` at the beginning of `Generator()` will return zero. The function `Normalize()` is called, which puts the terms in a canonical form, i.e. terms are ordered and collected with the correct coefficient. In our example, as the first fully substituted term we have *12, 1, 4, 6, 1, 1, 4, 6, 1, 1, 1, 3* before the call to `Normalize()`, which means we have a term $x * x$. `Normalize()` makes this into *8, 1, 4, 6, 2, 1, 1, 3*, which is x^2 .

Then, we loop over the statements in the compile buffer. `level` is the instruction counter. We have a long switch-clause that interprets the statement type identifiers like `TYPECOUNT`. Statements with `TYPEEXPRESSION` are not treated here. So we loop over all the compile buffer statements here and only call `TestMatch()` at the loop's end. This function has no effect in our example, because we have no pattern matching going on.

Then, the function `PutBracket()` is called to deal with brackets. Brackets are implemented by putting the special code `HAACKJE` inside the expression. The terms before the `HAACKJE` are outside the bracket, everything following it will be inside the bracket.

At the end of the loop over the terms in the expressions, the function `StoreTerm()` is called. This function puts the result of the processing in the output scratch buffers. Finally, we return to `Processor()`. There the final sorting is started. Also, the printing of the expressions is done here.

The parsing in `PreProcessor()` continues with

```

11
12      #do i=2,3
13      Id x?^'i' = x;
14      #enddo

```

Here we have a somewhat more complicated example of preprocessor instructions. The do-loop is treated in `DoDo()` which sets up data structures (`DOLLOOP`) to guide the preprocessor when it is parsing the loop body. The statement line will then be presented to the compiler two times and with the correct values of the preprocessor variable `i`. The compiler deals

with this statement in `CoId()` which is just calling `CoIdExpression()`. `CoIdExpression()` puts a `TYPEIDNEW` code into the lhs compile buffer. This tells the processor later how to do the pattern matching. The rhs is the term `x` that will be inserted.

The parsing continues and ends with

```
15
16      Print +s;
17      .end
```

The way these statements are treated and how the program is executed has already been described. The pattern matching is something that has not occurred before, though. We will not describe it here, since there is a dedicated section in this manual for that. After the final sorting, FORM will clean up temporary files and other resources that are not automatically freed by the operating system before FORM ends itself.

4 Specific topics

4.1 Pattern matching

to be written

4.2 The problem of dummy indices

FORM has a indices that can be automatically renumbered. With this we mean that when we have an expression like

$$f(i)*g(i)*h(j)*k(j)-f(j)*g(j)*h(i)*k(i)$$

we can say

Sum i,j;

and FORM will change the expression into

$$f(N1_?)*g(N1_?)*h(N2_?)*k(N2_?)-f(N2_?)*g(N2_?)*h(N1_?)*k(N1_?)$$

in which `Ni_?` are internal indices.

These internal indices follow a number of rules:

1. their numbers (AC.CurDum) start at AM.IndDum, which again starts at AM.DumInd+WILDOFFSET and AM.DumInd starts at AM.OffsetIndex + 2*WILDOFFSET. Hence AC.CurDum starts at AM.OffsetIndex + 3*WILDOFFSET. Because we need this extra space WILDOFFSET cannot be too large and this limits the number of indices that is allowed.
2. The dimension of the dummy indices is equal to the default dimension.
3. The internal (dummy) indices can be renamed at any time in order to create uniquely minimal terms. In the above expression that would mean that the second term would be 'rearranged' into

$$f(N2_?) * g(N2_?) * h(N1_?) * k(N1_?) \rightarrow f(N1_?) * g(N1_?) * h(N2_?) * k(N2_?)$$

and the expression becomes zero.

There are problems with this concept.

1. Multiplying expressions with dummy indices could give a repetition of the same indices as in $(f(N1_?) * g(N1_?))^3$. This has been solved partially as can be seen with the following program:

```
CF  f,g;
L   F = (f(N1_?)*g(N1_?))^3;
L   G = f(N1_?)*g(N1_?);
.sort
L   G3 = G^3;
Print;
.end
```

The routine that takes care of the proper shifts in dummy numbers is MoveDummies(). As one can see from the example, the SUBEXPRESSION to a power isn't treated this way. It would have a serious impact on the speed. With the G^3 it is different because that is slower to begin with.

2. Keep Brackets is extremely dangerous. The problem here is

$$f(N1_?) * (g(N1_?) * h(N2_?) * k(N2_?) + g(N2_?) * h(N1_?) * k(N2_?))$$

What is inside the brackets is invisible during the module. Hence a renumbering that involves `f(N1_?)` only can change `N1_?` into `N2_?` (FORM doesn't know there is already a `N2_?`) and anyway, the corresponding `N1_?` remains as it is. It means that there are complications with `Sum`, `Trace4` and things like `id p = f(?)`; which can generate dummy indices.

The second problem requires some action.

- A When `Keep Brackets` is active, renumbering should not be allowed, until the contents are multiplied with the outside of the brackets.
- B The multiplying with the contents of the bracket should follow the same procedure as the multiplication with a complete expression (`MoveDummies()`).
- C Introduction of new dummy indices should be above `AM.IndDum + WILDOFFSET/2`. These should vanish when the term is renumbered after multiplying the outside of the bracket with the inside.

`Trace4` involves the creation of dummy indices, but these vanish again without renumbering. Hence they don't cause problems.

In order to implement A-C we have to have a good look at all routines that use `AR.CurDum` and call `ReNumber()` or `DetCurDum()`.

4.3 Values of indices (and vectors)

The indices and vectors share common use. That means that vectors can occur in the places that are reserved for indices. In addition we have various types of indices. Hence it is important to know what range of values in an index location refers to what.

1. Special values:

<code>GAMMA1</code>	0	Dirac unit matrix
<code>GAMMA5</code>	-1	Dirac gamma 5 (only defined in 4 dimensions)
<code>GAMMA6</code>	-2	Dirac (1+gamma5) (only defined in 4 dimensions)
<code>GAMMA7</code>	-3	Dirac (1-gamma5) (only defined in 4 dimensions)

The above 4 indices are to be used only inside the function `g_`.

FUNNYVEC	-4	Used in <code>replace_</code> to indicate a vector with an unspecified index. Hence <code>VECTOR,4,numvec,FUNNYVEC</code> instead of <code>INDEX,3,numvec</code> .
FUNNYWILD	-5	Used to indicate an argument field wildcard like <code>?a</code> inside a tensor.
SUMMEDIND	-6	Used in <code>DELTA</code> to indicate <code>d_(mu,mu)</code> -4 as generated in traces.
NOINDEX	-7	Used by <code>ExecArg()</code> in splitting a multi-delta or multi-index. Taking out one to make a new argument we leave the old one with two or one empty spots.
FUNNYDOLLAR	-8	Used to indicate a dollar variable inside a tensor.
EMPTYINDEX	-9	Used in the bracket statement to indicate a <code>d_</code> . Because <code>d_</code> isn't a regular function we cannot use the function notation and it needs two arguments.
MINSPEC	-10	

MINSPEC must be smaller than all the other special values.

2. Fixed indices. They are in the range of 1 to `AM.OffsetIndex-1`.
3. Vectors are in the range from
`AM.OffsetVector = -2*WILDOFFSET+MINSPEC;`
to
`AM.OffsetVector + WILDOFFSET`
4. Wildcard vectors are in the range
`AM.OffsetVector + WILDOFFSET`
to
`AM.OffsetVector + 2*WILDOFFSET`
5. Regular indices are in the range from
`AM.OffsetIndex` to `AM.OffsetIndex + WILDOFFSET`
6. Wildcard indices are in the range
`AM.OffsetIndex + WILDOFFSET (=AM.WilInd)`
to
`AM.OffsetIndex + 2*WILDOFFSET (=AM.DumInd)`

7. Unused in the range of
`AM.OffsetIndex + 2*WILDOFFSET (=AM.DumInd)`
to
`AM.OffsetIndex + 3*WILDOFFSET (=AM.IndDum)`
8. Summed indices (`Ni.?`) are in the range of
`AM.OffsetIndex + 3*WILDOFFSET (=AM.IndDum)` to
`AM.OffsetIndex + 4*WILDOFFSET`
9. Unused in the range of
`AM.OffsetIndex + 4*WILDOFFSET`
to
`AM.OffsetIndex + 5*WILDOFFSET (=AM.mTraceDum)`
10. Summed indices as generated by the trace routines are above
`AM.OffsetIndex + 5*WILDOFFSET (=AM.mTraceDum)`

Note (JV): I am not sure why there are unused regions. I must have had a reason for them, but I have forgotten about it (it was more than 20 years ago). And then, maybe it is used somewhere in a totally untransparent way.

Note 2 (JV): It was good to make this list. It turned out that in several places the code that checks for wildcard indices was only limited from below, not from above. It would of course be very rare to run into trouble with this, but it is better to have the code formally correct. One never knows. This was particularly the case in `FindRest()` (in `findpat.c`). There may be more. It is best to repair this, whenever encountered.

From the above it should be clear that on a 32-bits computer
`5*WILDOFFSET+AM.OffsetIndex+nTraceDummies < 215`
in which `nTraceDummies` is the number of dummies that can be introduced when taking a 4-dimensional trace.

If we assume that we will not take traces of more than 200 gamma matrices (each with a different index, because otherwise there are contractions) `nTraceDummies` will be at most 100. `AM.OffsetIndex` is by default 128. The value that we selected for `WILDOFFSET` is 6100 which allows a maximum value of 2167 for `AM.OffsetIndex`.

5 The test suite

The subdirectory `check` contains a test suite for FORM. Using the autoconf facilities the checks can be started with the command `make check`. Otherwise, one can issue the command `ruby form.rb` in the `check` directory.

The test suite is written in the language Ruby¹. Ruby itself already offers a unit testing framework and this is used with as minimal as possible extensions to make the creation of test cases for FORM programs easy. All the extensions to the built-in Ruby testing framework (`Test::Unit`) are contained in the file `form.rb`. This file also contains code to load test cases from other `*.rb` files in the `check` directory. Therefore all test cases are contained in appropriately named `*.rb` files. The makefile's purpose is to integrate the call `ruby form.rb` into the autoconf system.

Side note: The choice to use Ruby and its built-in test framework was taken for several reasons: It makes sense to use or adapt already existing testing frameworks in order to keep the extra cost of maintenance as low as possible for the FORM programmers. There are numerous systems available on the market, some are part of a language runtime environment (libraries), and some are dedicated programs with a custom configuration language. Since the tests for FORM programs center mainly about text processing, i.e. comparing the textual FORM output to a correct answer, we need powerful text processing facilities like pattern matching. But we also need file operations and information from the operating system to check the run of a FORM program, eventually. All this is readily available in the testing frameworks of scripting languages, like Ruby, Python, or Tcl. Ruby was ultimately chosen, because the mixing of FORM code with the steering scripting language code looked nicest, and the small amount of extra (Ruby) syntax necessary makes it convenient to add new test cases.

A new test case can be implemented in the following way. First of all, we need a FORM program that is to be run. It might be a program that exhibits an actual bug in (a previous version of) FORM or that contains generic code that should be guaranteed to work, also in coming releases of FORM. It might also be code that deliberately crashes FORM or causes other errors, like syntax errors, if this behavior of FORM is to be assumed. Usually, the FORM program is rather short or can be made such. In this case, we are going to mix the Ruby and the FORM code in one file. Alternatively, the FORM program can also be kept in a separate file. This option will be discussed later.

Now, either one chooses an existing `*.rb` file (not `form.rb`) or starts a new one. The name of the file should fit the test case scenario. In this file we need to define a Ruby class that will contain our FORM code as well as the checks (assertions) we want to impose on the run.

The generic frame of this test case definition looks like this:

¹<http://www.ruby-lang.org>

```

class [Test name] < FormTest
def setup
  [Setup code, usually this includes the FORM program code]
end
def test1
  [Execution code, and the assertion and testing code]
end
end

```

The text in the brackets [] needs to be filled with our specific code. The details of the Ruby code itself will be explained later. For a start, it is usually advisable just to copy an existing test case and modify it.

Every class defined in this way will be used for the testing. First, Ruby will run the code in the class method `setup`, and then it runs `test1`.

A complete test might look like this:

```

class SymbolIdTest < FormTest
def setup
  input <<-EOF
S x, y;
L f = (x+y)^100;
id x = y;
print;
.end
  EOF
end
def test1
  execute FORM
  assert no_problem
  assert result("f") =~
    pattern("1267650600228229401496703205376*y^100;")
end
end

```

We have chose the name `SymbolIdTest` for our class. We defined the FORM program in-line with a so called here document (`<<-EOF ... EOF`). We do run the FORM executable. Alternatives would be `TFORM`, for example. The assertions we have are that no problem occurred, i.e. no syntax error, no runtime error, or similar things. We also check the output of our FORM program. We compare via pattern matching the result of the

expression `f` with the correct answer. The function `result()` extracts the appropriate line from the output, `=~` is the pattern matching operator in Ruby, and the function `pattern()` prepares special characters like the caret (^) for the pattern matcher.

Next time we run the test suite, our test will be run as well. If no assertions are violated, we will only see the number of successful tests and assertions increased in the summary output.

Even though the extra Ruby syntax is kept to a minimum and is rather straightforward, some remarks about the Ruby language are useful here. Classes are defined by the keyword `class`, and methods (or functions) are declared with the keyword `def`. These definitions are always ended with the keyword `end`. `FormTest` is a class defined in `form.rb` that contains all the special code for FORM test and that is derived from the built-in Ruby test case class `TestCase`. For every test case we derive again from this class (`class B < A` says that `B` is derived from `A`). We don't need semicolons to end a line and indentation is arbitrary. Class names should be capitalized. In Ruby, parentheses around the arguments of functions can often be omitted. We use this possibility when we call the functions `input`, `execute`, and `assert`. We could have written `execute(FORM)` as well, for example. The here document (`<<-EOF ... EOF`) can also use other markers instead of `EOF`, of course. The minus sign before `EOF` allows the end marker to be indented. Comments are started with a `#`.

One class can actually contain more than one test. The testing framework will call the method `setup` and then a method whose name starts with `test` (Note: in newer versions of Ruby the name could be just `test`, but older versions ($\geq 1.8.x$) require at least one following extra character). If there are more methods starting with `test`, each will be called and for each `setup` will be called first.

In `setup` we need to prepare everything for the execution of FORM. We can either use `input` to in-line the source directly, or we can use `input_file` with a string as an argument to reference an external file, e.g.

```
input_file "parsebug.frm"
```

The function `input` will create a temporary FORM file for the contents. The name of the file is defined in `form.rb`. The executable will later be run with the given name or the name of the temporary file as an argument. If additional arguments need to be given to the executable, the function `extra_parameter` can be used, like e.g.

```
extra_parameter "-w4 -l"
```

Sometimes one might need to prepare more things for a FORM run, like setting up certain files or starting an external program. This needs to be done by ordinary Ruby code. For this, some more of the Ruby language needs to be known by the user.

In the class methods with a name starting with `test` we put the code to run the FORM executable and to test the outcome. Usually, the first line will be the call to the executable itself, either

```
execute FORM
```

or

```
execute TFORM
```

(PARFORM is not supported yet). The function `execute` will run the executable with the necessary or requested arguments, but it will run it under the supervision of the `strace` system utility. Therefore `strace` needs to be present on the system (options to enable or disable the use of `strace` will probably be added in the future). `strace` is used to get detailed information about the return value or possible failure states of the executable. The output of `strace` will be saved in a temporary file and made available to the test case programmer in a Ruby variable. The regular output and the error channel output will be available in Ruby variables as well.

The Ruby variables containing the output are `@strace_out`, `@stdout`, and `@stderr` (the leading `@`-sign is Ruby syntax for specifying instance variables, i.e. variables belonging to a certain object). These variables are the primary source for doing tests. In principle, these variables can be investigated directly, for example via pattern matching like

```
if @strace_out =~ /Segmentation fault/
  ...
end
```

which checks whether a segmentation fault has occurred (the slashes in Ruby define a pattern). But for the most common cases some test functions exist that encapsulate necessary pattern matching details. These functions return true or false values which can be used as arguments to the `assert` function. The `assert` function raises an error if the argument is false.

Available tests functions are:

<code>crash</code>	true if a crash (segmentation fault) occurred
<code>warning</code>	true if FORM has issued a warning
<code>compile_error</code>	true if FORM has found a syntax error
<code>runtime_error</code>	true if FORM has terminated prematurely
<code>error</code>	true if <code>compile_error</code> or <code>runtime_error</code> is true or the standard error channel contains data
<code>problem</code>	true if <code>warning</code> or <code>error</code> or <code>crash</code> is true

Additionally, the logical opposite of each function exists with a name starting with `no_`, like `no_problem` or `no_crash`.

There is also the function `return_value` which gives the return value of the FORM program as an integer, so one could do a check like

```
assert return_value == 66
```

If pattern matching is coded directly, like in our example, some details have to be considered. The operator `=~` will try to match a string with a pattern. The variables like `@stdout` are actually strings (they do contain the carriage return and/or line feed for multi-line output). Patterns in Ruby are written between slashes and various characters are interpreted in a special way (following the widely used regex-syntax).

There are four functions to facilitate things: `result()`, `pattern()`, `exact_result()`, and `exact_pattern()`. `result()` takes a string being the name of an expression and returns a string that only contains the lines belonging to the last output of this expression. If it is not the last output of an expression that is wished for, a second numeric parameter can be given that specifies the index of the output (counting starts at 0). While `result()` removes all line breaks and whitespaces, `exact_result()` leaves them in place. `pattern()` transforms special characters in the given string, removes whitespaces and line breaks, and returns the string as a pattern. Since FORM expressions usually contain a lot of special characters like `+`, `*`, `.`, etc. they cannot not be simply used in a pattern. `pattern()` transforms these characters automatically into the correct regex equivalent, e.g. `+` becomes `\+`. With it, a FORM expression can be directly given as an argument and used in a pattern matching (see example). `exact_pattern()` does not treat whitespaces and line breaks in a special way as `pattern()` does and can therefore be used when a exact comparison is required (if for example a bug in the output functions of FORM had caused some whitespace or line breaks to be missing and a test case were required to check for this behavior).

If one doesn't want or cannot use the `assert` function, one can signal a test failure to the testing framework by raising an `AssertionFailedError` directly, like for example

```
if return_value != 2
  raise AssertionFailedError.new("return value is wrong!")
end
```

Suppose a FORM program should have deleted some file (`#remove`), one could implement the following test

```
if File.exist?("thenameofthefile")
  raise AssertionFailedError.new("File still exists!")
end
```

The testing framework actually not only calls `setup` and each `test` method but also a method called `teardown`. This method is responsible for cleaning up things at the end of each test run. The class `FormTest` provides such a `teardown` method that will be inherited by the users test case class unless it is overwritten. It calls the method `remove_files` to delete all temporary files that have been created so far. `remove_files` can be called by the user directly. If `teardown` is to be replaced by a specific implementation, it is advisable to still call `FormTest`'s `teardown` (using Ruby's command `super`), like for example

```
...
def teardown
  super
  File.delete("extra.log")
end
...
```

At last, a complete example as it is actually contained in the repository.

```
#[ SparseTable1 :
=begin
  Bugs reported 2004-04-06 by Misha Tentukov
  PrintTable and FillExpression did not work with non-sparse tables
  Fixed 2005-09-27
=end
class SparseTable1 < FormTest
def setup
  input <<-EOF
cf f;
s x;
ctable Tab(1:'TableSize');
ctable TabNew(1:'TableSize');
```

```

#do i=1,'TableSize',1
Fill Tab('i')=f('i');
.sort
#enddo

* BUG1 (not all elements are printed):
PrintTable Tab;

bracket x;
.sort
L expr1=table_(Tab,x);
print;
.sort

bracket x;
.sort

* BUG 2 ( seems only TabNew(1) is ok - further everything is broken):
Fillexpression TabNew=expr1(x);
.sort

#do i=1,'TableSize'
L e'i'=TabNew('i');
#enddo
print;
.sort
.end
EOF
extra_parameter "-D TableSize=10"
end
def test1
execute FORM
assert no_problem
assert result("expr1") =~ pattern(<<-EOF
f(1)*x + f(2)*x^2 + f(3)*x^3 + f(4)*x^4 + f(5)*x^5 + f(6)*x^6 + f(7)*x^7
+ f(8)*x^8 + f(9)*x^9 + f(10)*x^10;
EOF
)
assert result("e10") =~ /\s+f\10\;/
end
end
#] SparseTable1 :

```

Some remarks. Folds are used (to structure a long file). `=begin` and `=end` define a commentary block. Here useful information are given about the bug that triggered the test case. The input is not modified compared to the original FORM program, it is just directly pasted into this Ruby file. We use `extra_parameter` to define a preprocessor variable for the run. We check `expr1` to a multi-line reference. Since we use `pattern()` (instead of `exact_pattern()`), we can be sloppy about the indentation and the whitespaces. The expression `e10` is matched to a pattern done "by hand" instead (just to show the principle). For such a test case, where

we are mostly interested about the correctness of the calculation, the first assertion (`assert no_problem`) is a standard.

6 CVS

The CVS repository resides in `/user/form/cvs_repository`. It is advisable to set the environment variable `CVSROOT` accordingly, like (using bash shell syntax)

```
export CVSROOT=:ext:myusername@mytrustedmachine.nikhef.nl:/user/form/cvs_repository
```

A mailing list exists for CVS commits. The administration interface for this mailing list can be found under the web address

```
https://mailman.nikhef.nl/cgi-bin/admin/form-cvs
```

A password is required.

Click *Membership Management* and then *Mass Subscription* to add new people. The personal details of the subscribers like the email address or the name can be changed under *Membership Management* as well.

The triggering of the CVS commits mails is done in the following way. In the file `loginfo` in the directory `CVSROOT` (inside the repository) the default action for logging is set such that the script `/user/form/cvs-log.sh` will be called with the committer's user name and the CVS mailing list user name. The shell script does some simple message transformation and then uses the command `mail` to send the commit mail to the mailing list.

6.1 Some useful CVS idioms

To just show what would be updated/changed without actually modifying anything, use

```
cvs -n update
```

If `cvs -n update` has shown you that something new in the repository will be merged into your directory and you want to know in advance what the details are, you can do for each of the files involved a

```
cvs status <filename>
```

and note the version number of your local file, and then do a

```
cvs diff -r <versionnumber> <filename>
```

to see the differences.

In case you want to compile an older version of FORM (maybe to find out whether a certain bug is already present or not), do

```
cvcs update -D "<DATE>"
```

to checkout the sources as they were on a certain date, e.g.

`cvcs update -D "2006-05-12"`. The files will get the so-called sticky flag, which do prevent simple `cvcs update` commands in the future to update to the latest version from the repository. To remove the sticky flag on a file use

```
cvcs update -A <filename>
```

Without the filename all files will have the sticky flag removed.